

12th Feb 2026 / CS6370

Assignment
+
Written Test

→ candidate generaⁿ
 $\left(\frac{n_c}{N}\right)$

bayesian approach of how pharmacists
understand doc's handwriting

- smoothing

{a, b, c, d, e}

↓ ↓

present

in dictionary but not in
corpus

no. of types = 5
no. of tokens = 7

a
b

a
b

a
b c

	F	RF
a	3	3/7
b	3	3/7
c	1	1/7
d	0	0
e	0	0

data Mining → David Hand

ATZ

add-1 smoothing → just add 1 to each
(laplace) frequency. We will
get a revised PMF. In this
way, we have received a non-0 man.

$$\frac{n_c}{N} \rightarrow \frac{n_c + 1}{N + 1}$$

- candidate corrections

deletion, insertion, transposition (ac → ca), substitution

diff. transformations are possible.
We can give probab. to each of these transformations.

bottom-up
approach

$\text{del}(x, y)$, $\text{ins}(x, y)$, $\text{trans}(x, y)$, $\text{sub}(x, y)$

this works only with
single letter mistakes

$$p(t|c) = \begin{cases} \frac{\text{del}(x, y)}{\text{count}} \\ \vdots \end{cases}$$

substitution of
 x (incorrect) for
 y (correct)

- parallel corpus → is a collecⁿ of texts in 2 or more lang. that are direct translations of each other, aligned sentence-to-sentence
- parse trees ^(syntax tree) → tree-shaped diagrams that show the grammatical structure of a sentence or expression acc. to set of grammar rules.
- evaluate the candidates

$$c \quad \text{freq}(c) \quad p(c) \quad p(t|c) \quad p(t|c)p(c) \quad \%$$

first find the type of error & then calculate $p(t|c)$.

- currently we are doing localizaⁿ. This wouldn't have been the case if there were words at multiple positions.
- we take the word (suggestion) with highest percentage.

acres → 'acress'

(s → es) insertion

acress → acress

(s → ss) insertion

these
&
are
totally
diff.

but it can be added.

- all good, we can change the word, that is dependent on the context.

$P(\text{acress})$ $\underbrace{c_5 \dots c_5}_{\text{same context words}}$ evidence and then calculate

this is our likelihood term that this is context word given candidate word. → its probability